Science and Technology

10. Tissues and Its Types

TISSUE: Cells combine to perform same function.

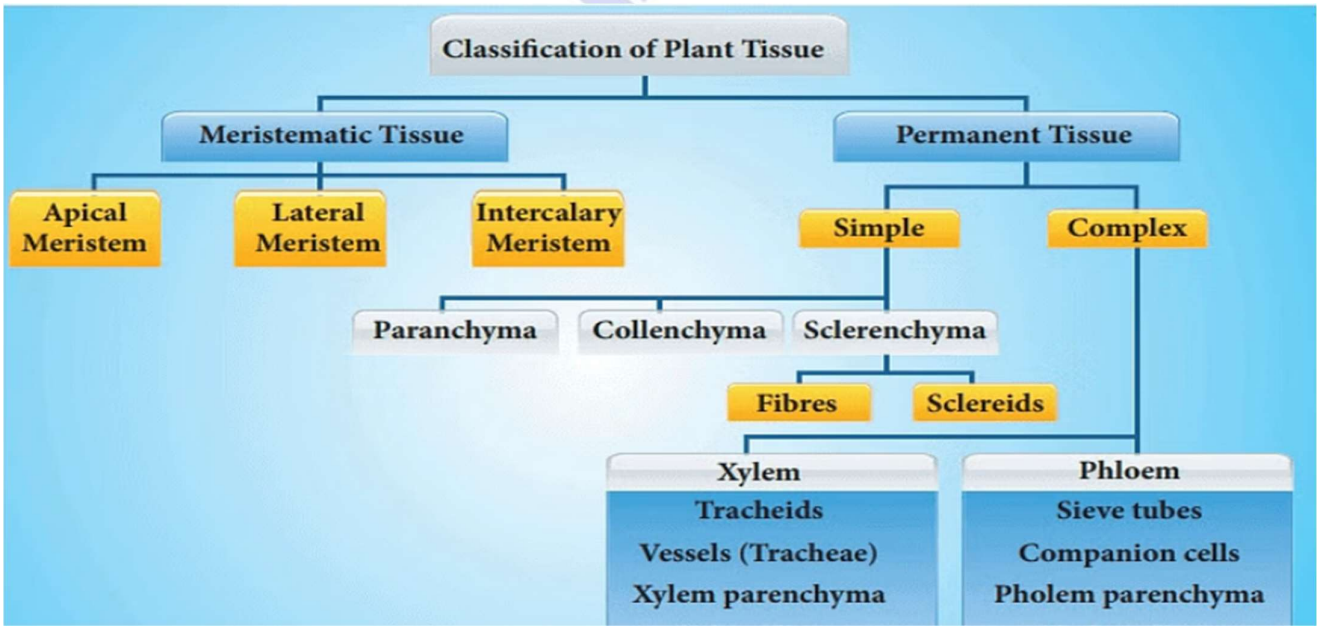
- When plant cells combine, they form plant tissues.
- When animal cells combine, they form animal tissues.
- Causes division of labour.
- Decreases workload on individual cells, hence leads to organisation.

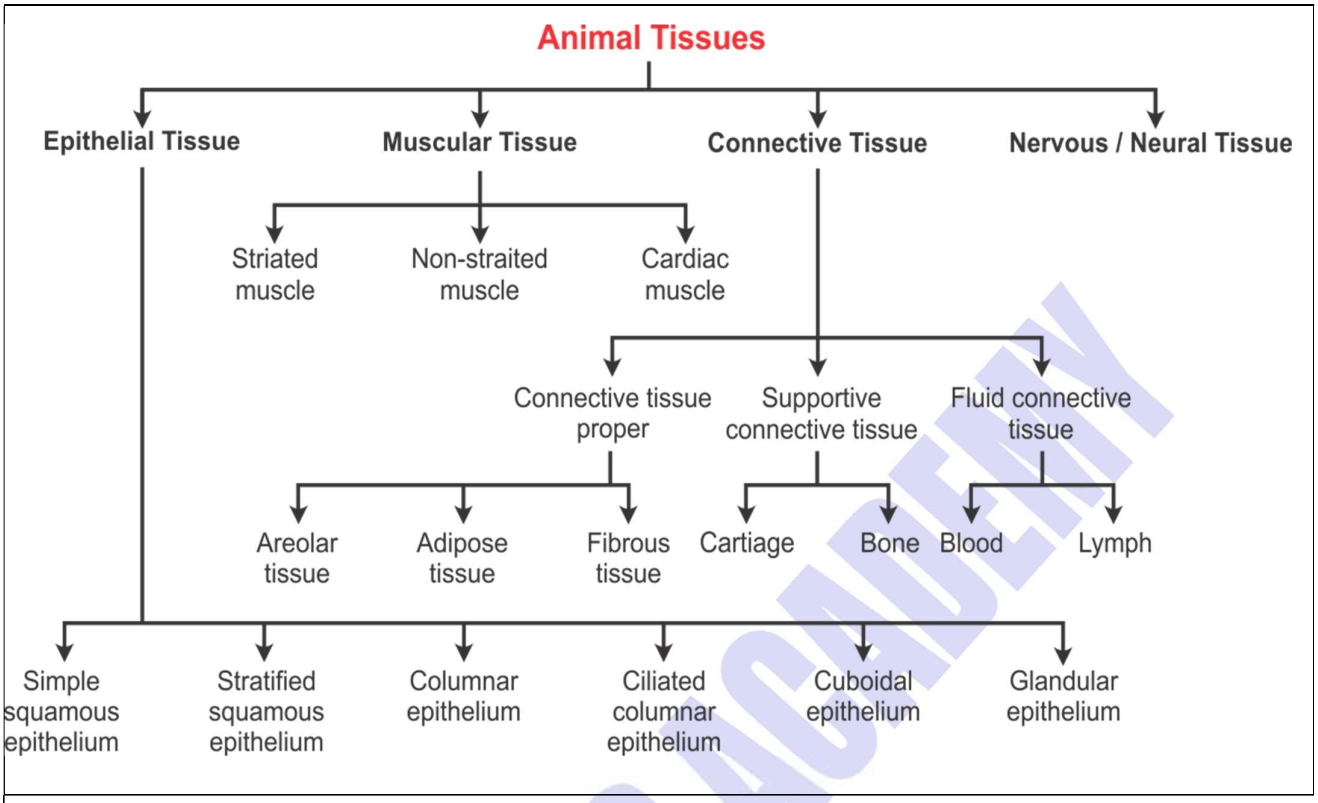
Cell→ Tissues→ Organ→ Organ system→ Organisms

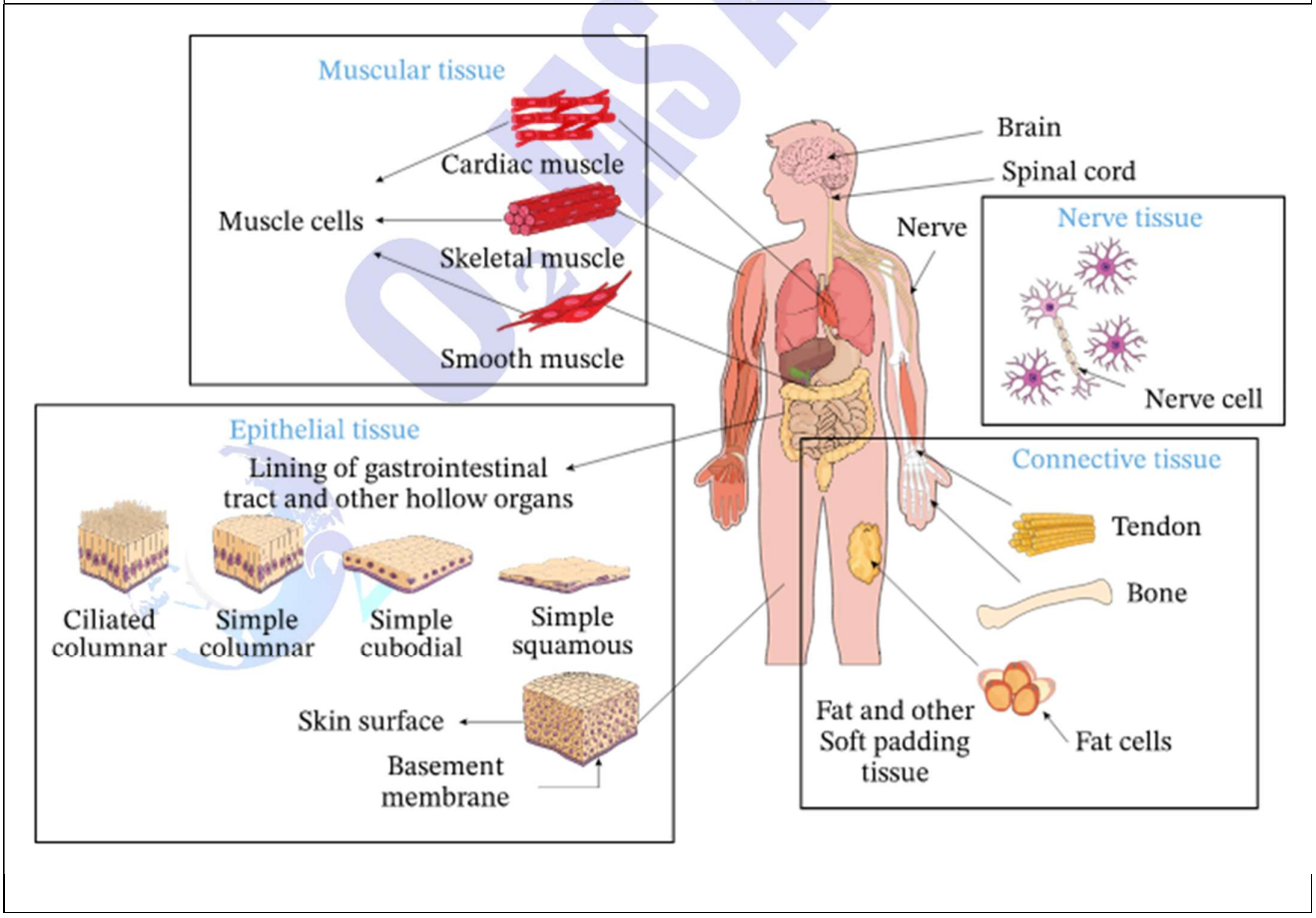
Mitosis	Meiosis
1) It occurs in somatic cells.	1) It occurs in germ cells.
2) Nucleus divides only once.	2) Nucleus divides twice.
3) Two daughter cells are formed.	3) Four daughter cells are formed.
4) Daughter cells are diploid.	4) Daughter cells are haploid.
5) It occurs more frequently.	5) It occurs less frequently.
6) Daughter cells form somatic organs.	6) Daughter cells form gametes.
7) There is only one prophase, one metaphase one anaphase and one telophase.	7) There are two of each phase and five sub-phases in prophase - 1.
8) Number of chromosomes are not changed in the daughter cells.	8) Number of chromosomes are reduced to half.
9) Chromosome number doubles at the beginning of each cell division.	9) Chromosome number is not doubled. It doubles after the end of first meiotic division.
10) No crossing over in chromosomes.	10) Crossing over occurs chromosomes.
11) Equation division.	11) Reduction division.

Table 7.3: Difference between mitosis in Plants and Animals

Plants	Animals
Centrioles are absent	Centrioles are present
Asters are not formed	Asters are formed
Cell division involves formation of a cell plate	Cell division involves furrowing and cleavage of cytoplasm
Occurs mainly at meristem	Occurs in tissues throughout the body







O2IAS ACADEMY

There are mainly four different types of tissues present in our body.

1. **Epithelial-** provides covering or lining
2. **Muscular**- helps in movement
3. **Neural**- responds to stimuli
4. **Connective-** supports, links and cushions

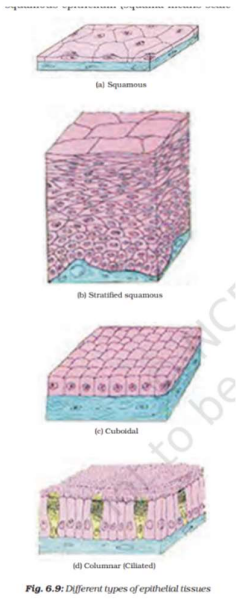
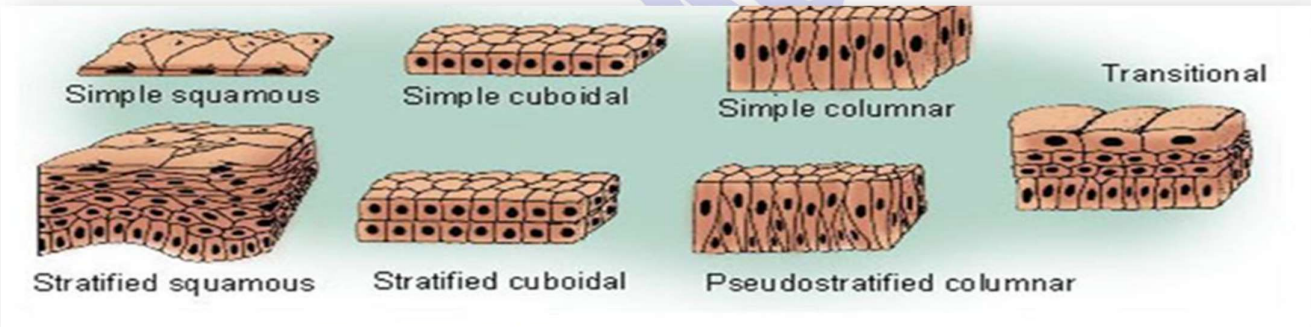


Fig. 6.9: Different types of epithelial tissues

- Epithelium covers most organs and cavities within the body.
- It also forms a barrier to keep different body systems separate.
- Epithelial tissue cells are tightly packed and form a continuous sheet.
- They have only a small amount of cementing material between them and almost no intercellular spaces.
- All epithelium is usually separated from the underlying tissue by an extracellular fibrous basement membrane.
- Epithelial cells often acquire additional specialisation as gland cells, which can secrete substances at the epithelial surface. Sometimes a portion of the epithelial tissue folds inward, and a multicellular gland is formed. This is glandular epithelium.

Types of Epithelial Tissue	Location	Structure	Function
Simple squamous	Blood vessel lining, capillaries, air sac lining of lungs, Bowman's capsule of kidney	A single layer of flat cells having irregular boundaries	Transport by diffusion and where minimal protection is required
Simple Cuboidal Epithelium	The tubular lining of kidneys, glandular ducts	A single layer of short cylindrical cells. It may have **microvilli as in proximal convoluted tubules**	Absorption and secretion
Simple Columnar Epithelium	Digestive tract and upper respiratory tract lining	A single layer of columnar cells (tall and slender) and often ciliated	Protection, absorption, mucus secretion and movement in a specific direction
Stratified Squamous Epithelium	The lining of the mouth and vagina	Made up of several layers of cells, continuously sloughed off and regenerated. The older layer of cells is pushed upwards and becomes flat. The lower layer is columnar and metabolically active	Protection
Stratified Cuboidal	Mammary glands, sweat gland and salivary glands	The upper layer is cuboid and other layers may be cuboidal or other types	Protection of ducts of various glands
Stratified Columnar	Male urethra and lobar ducts of salivary glands	There is a layer of columnar cells present on squamous, columnar or cuboidal epithelial cells	Protection and secretion

Pseudo-stratified Columnar	Respiratory passage and ducts of many glands	Similar to columnar epithelium but all the cells are not of similar height	Protection, secretion and movement of mucous
Keratinised	The outer or apical layer of the cell	Mostly dead and devoid of nucleus and cytoplasm. The cytoplasm gets replaced by keratin, which makes the layer waterproof	Protection against abrasion
Transitional epithelia or urothelium	Urinary bladder, urethra, ureter	Stratified epithelium, which can contract or expand as per the requirement. Cells are cuboidal when not stretched, but when the organ stretches, then tissue gets compressed and cells appear irregular and squamous-shaped	Stretch readily to accommodate the different volume of liquids Act as a barrier and have tight junctions to prevent reabsorption of toxic substances

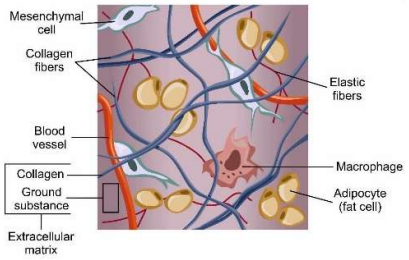


Connective Tissues

- Connects body parts
- Prevents organs from getting displaced by body movement.
- Acts like binding, supporting and packing material.
- Cells are loosely spaced.
- Cells are embedded in an intercellular matrix.
- Matrix may be jelly, fluid, dense or rigid.

- Connective tissues, support and connect different tissues and organs of the body.
- **Connective tissue is made up of :-**

- Few cells present in the intercellular framework
- The cells also secrete a thin gel of polysaccharides, which together with fibres make matrix or ground substance.
- Protein fibres secreted by the cells, known as collagen or elastin. The elasticity, flexibility and strength of the connective tissues are due to fibres.



- Collagen fibres are the most widespread and made up of fibrous protein, collagen. Collagen fibres are flexible and have high tensile strength.
- Elastic fibres form a network and can be stretched like a rubber band. They are made up of protein elastin. They retain their original shape and size once the force is removed.
- Reticulate fibres consist of collagen and glycoproteins. They are thin and form a delicate network. They join connective tissues to neighbouring tissues.

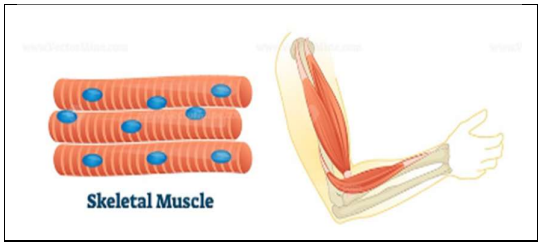
MUSCLE TISSUE/MUSCULAR TISSUE

Characterstics:

- Composed of muscle cell/ muscle fibres.
- Muscle fibres are elongated and large sized.
- Contractile proteins are present in fibres.
- Contraction and relaxation.

Types:

a. Skeletal
b. Smooth
c. Cardiac



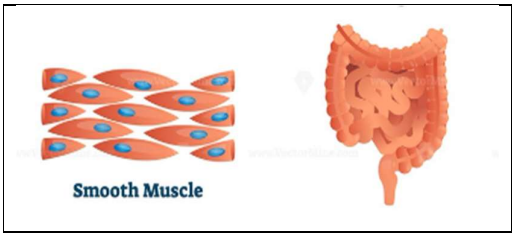
a. Skeletal Muscle Tissue:

Characteristics

- Muscle fibres are elongated and cylindrical
- Unbranched
- Attached to bones
- Each cell is multi-nucleated→ many nuclei.
- Striations are present → alternate dark and light bands.
- Easily fatigue
- Helps in body movement.
- Maintains posture

Other names: Skeleton muscle, Striated muscle and voluntary muscle.

b. Visceral Muscle/ Smooth muscle tissue



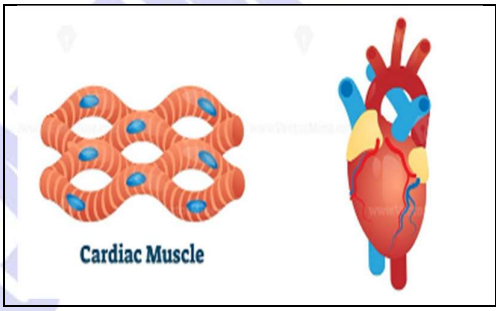
Characteristics:

- Fibres are spindle shaped/fusiform i.e. pointed at ends/tapering ends.
- Single and centrally located nucleus in each cell.
- Striations are absent.
- Unbranched
- Do not fatigue
- Performs functions like peristalsis.

Other names: Unstriated/Involuntary muscles

Example: Stomach, Intestine, eye, blood vessel etc.

c. Cardiac muscle tissue



Characters:

- Fibres are cylindrical in shape.
- Each fibre is uninucleate
- Fibres are branched
- Striations are present.
- Do not fatigue.
- Performs contraction and relaxation in wall of heart.
- Helps to pump and distribute blood to body.

Other names: Striated, Involuntary.

Three Types of Muscular Tissue

	Location	Function	Appearance	Control
Skeletal 	skeleton	movement, heat, posture	**striated**, multi-nucleated (eccentric), fibers parallel	**voluntary**
Cardiac 	heart	pump blood continuously	**striated**, one central nucleus	**involuntary**
Visceral (smooth muscle) 	G.I. tract, uterus, eye, blood vessels	Peristalsis, blood pressure, pupil size, erects hairs	**no striations**, one central nucleus	**involuntary**

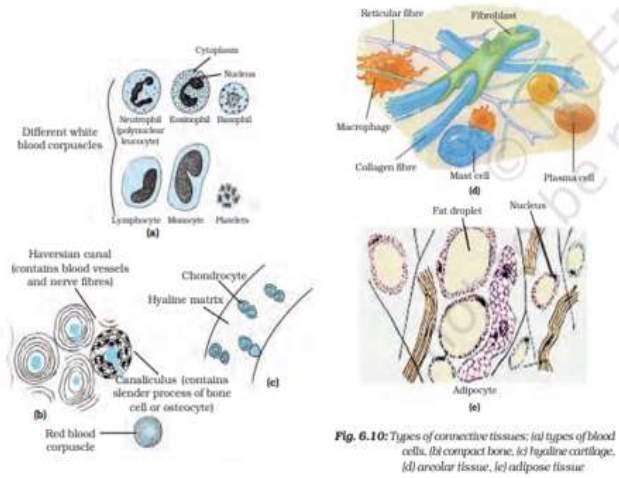
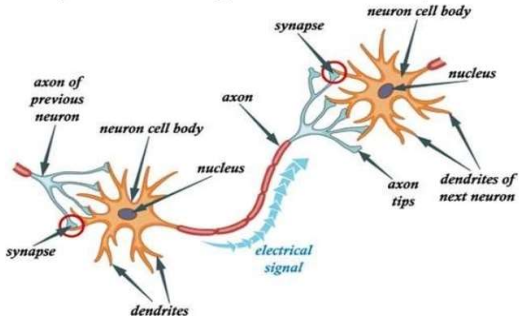


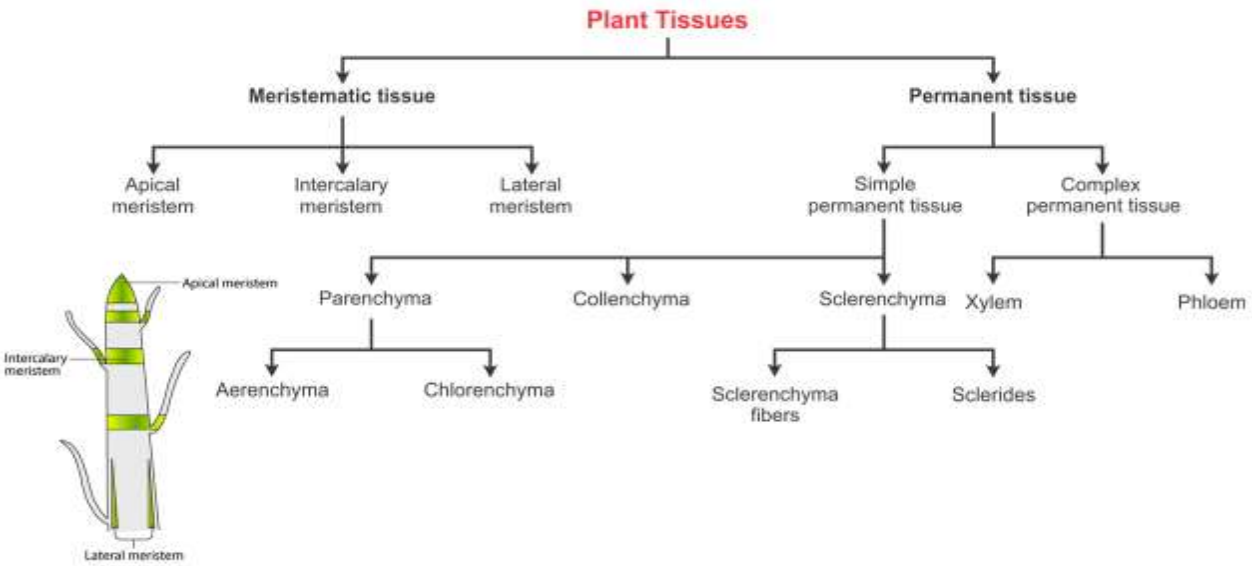
Fig. 6.10: Types of connective tissues: (a) types of blood cells, (b) compact bone, (c) hyaline cartilage, (d) areolar tissue, (e) adipose tissue

How do neurons transmit impulses (send message to other cells)?



Types of Connective Tissues

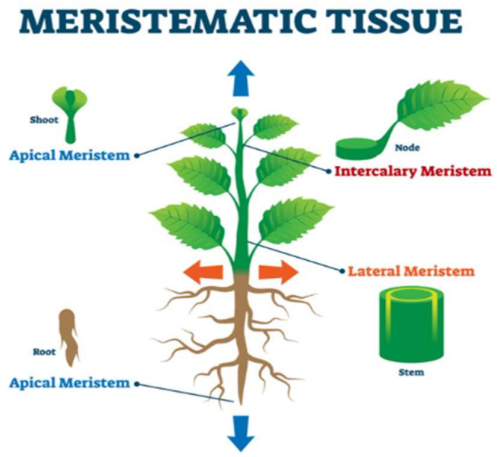
Type of Tissue	Location	Function
• Loose connective tissue	Deep layers of skin, blood vessels, nerves and body organs.	Support and elasticity
• Dense connective tissue	Tendons and ligaments	Compactly pack structures to one another and provides great strength.
• Elastic connective tissue	Lungs, arteries, trachea and vocal cords.	Provides elasticity
• Specialised connective tissue (i) Cartilage	Ends of long bones tip of nose, parts of larynx and trachea.	Provides flexibility and support.
(ii) Bone	Bones	Protection and support muscle attachment.
(iii) Vascular connective tissue (blood)	Within blood vessels	Transport of oxygen and carbon dioxide and immune response blood clotting.
• Adipose tissue	Deep layers of skin, surrounding heart and kidneys and padding around joints.	Support, protection, heat conservation and energy source.

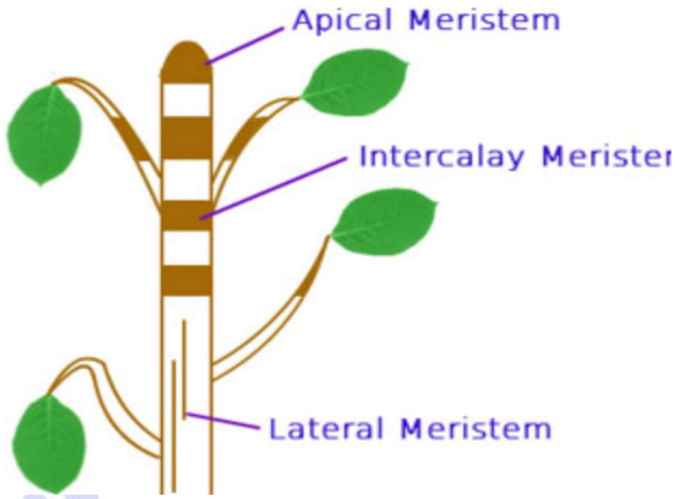


PLANT TISSUE

1. **MERISTEMATIC TISSUE:**

- **OCCURRENCE:** Growing region of plant like root, shoot etc.
- **Shape:** Spherical, oval/rectangular.
- Intercellular spaces: Absent, hence compactly packed.
- Vacuoles absent in it.
- Focus only on cell division.
- Highly active cells.





1. APICAL MERISTEM

- Present at root and shoot tip.
- Length of root and shoot increases.

2. LATERAL MERISTEM

- Present beneath bark.
- Girth of root or stem increases.

3. INTERCALARY MERISTEM

- Present at base of leaves or internodes.
- Length of organs like leaves and nodes increases.

2. PERMANENT TISSUES

- Cell looses ability to divide.
- Take up specific role.
- Thus are permanent.
- Differentiation: Process of taking up permanent shape, size, and function.
- Large central vacuole.
- Cell wall is thick.
- Helps in protection, support, storage and Photosynthesis.

SIMPLE PERMANENT TISSUE

- Cells are structurally and functionally similar
- One type of cells.
- Main function is to store food and perform photosynthesis.

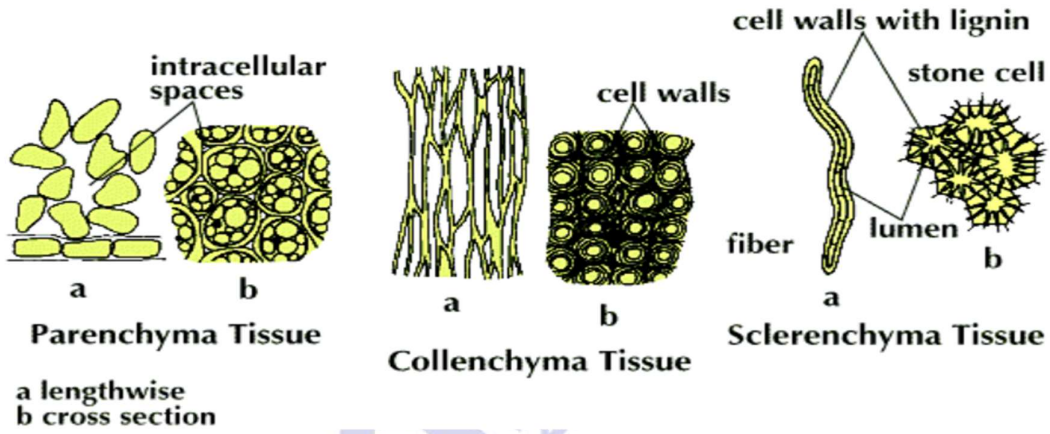
Types of Simple Permanent Tissues

1. **Parenchyma tissue:**

- Occurrence: Stem, roots, leaves, flowers and fruits.
- Cell structure: Thin cell wall, round polygonal or elongated in shape.
- Intercellular spaces are present.

Functions:

- Food storage tissue
- In some cases→ it contains chlorophyll, thus called **CHLORENCHYMA.**
- When loosely packed→ Intercellular spaces are present, called **AERENCHYMA**: provides buoyancy to parts and helps them to flow.



2. Collenchyma Tissue:

Occurrence: In leaf stalks and stems of dicots.

Structure of cells:

- Living
- Elongated
- Irregularly thickened at corners.
- Very little intercellular space.

Functions:

- ➢ Provides mechanical strength
- ➢ Provides flexibility and photosynthesis
- ➢ Allowed bending of parts.

E.g. Tendrils and stems of climbers

3. Sclerenchyma Tissue

Occurrence:

- In stems around vascular bundles.
- In veins of leaves.
- Hard covering of seeds and nuts.

Eg. Husk of coconut.

Structure:

- Long and narrow
- Walls are thickened by LIGNINS
- Cell cavity is absent due to excessive thickness.
- Intercellular spaces absent.
- Cells are dead.

Function:

- Provide strength to plant cell.
- Provides protection and mechanical strength

COMPLEX PERMANENT TISSUE

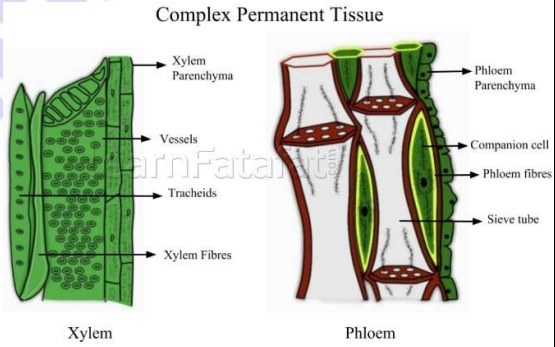
- Made up of more than one type of cells.
- Cells co-ordinate to perform common function.

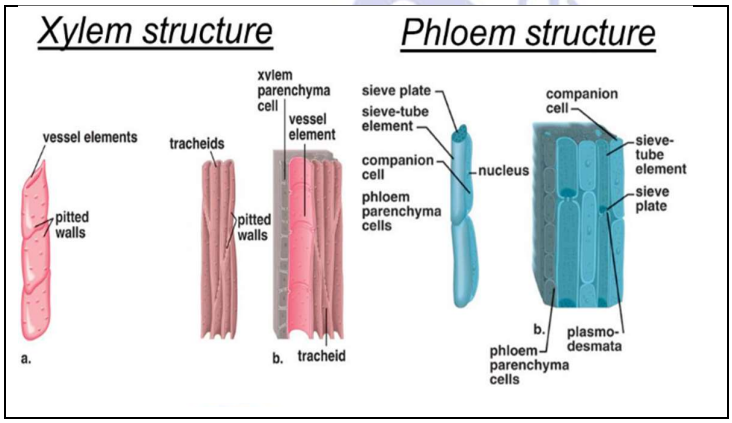
Functions:

- Transport water, minerals and food materials.

Two types:

1. Xylem
2. Phloem

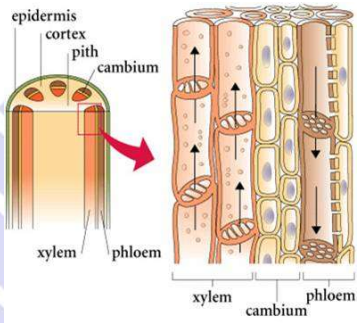




- Xylem consists of tracheids, vessels, xylem parenchyma and xylem fibres.
- Tracheids and vessels have thick walls, and many are dead cells when mature.
- Tracheids and vessels are tubular structures. This allows them to transport water and minerals vertically.
- The parenchyma stores food and help in sideways conduction of water.
- Xylem fibres are mainly supportive in function.

- Phloem is made up of five types of cells: sieve cells, sieve tubes, companion cells, phloem fibres and the phloem parenchyma.
- Sieve tubes are tubular cells with perforated walls.
- Phloem transports food from leaves to other parts of the plant.

S.No.	Xylem	Phloem
1.	It carries water and minerals from roots to leaves.	It carries food materials from the leaves to the growing parts of the plant.
2.	It is composed of dead cells.	It is composed of living cells.
3.	It has a thick cell wall.	It has a thin cell wall.
4.	Its cells are impermeable to the water.	Its cells are permeable to food components.
5.	Its cells are made up of lignin.	Its cells are made up of cellulose.
6.	It gives mechanical strength to the plants.	Phloem does not give mechanical strength



- Except phloem fibres, other phloem cells are living cells.

Q. Fat present below the skin surface in our body, acts as a barrier against: (HPAS 2020)

a. Loss of heat from our body
b. Loss of essential body nutrients
c. Loss of salts from the body
d. Entry of harmful micro-organisms from the environments

PRACTICE QUESTIONS

Q. Consider the following statements:

	Tissue	Location
1.	Apical	Present at root tips only
2.	Lateral	Present beneath bark
3.	Intercalary	Present at base of leaves or internodes

Choose the correct statement(s):

a. 1 only
b. 1 and 2.
c. 2 and 3.
d. All are correct

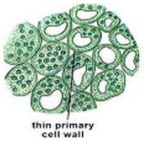
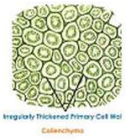
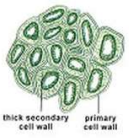
Q. Consider the following statement(s):

Tissue **Function and constituents**

1. Parenchyma Large vacuole and store food
2. Sclerenchyma In climber plants and provide mechanical strength
3. Collenchyma contain lignin

Choose the correct option:

a. 1 only.
b. 2 only.
c. 1 and 3 only.
d. All are incorrect.

Features	Parenchyma	Collenchyma	Sclerenchyma
			
1. Cell shape	Isodiametric cells which are oval, spherical or polygonal in shape	Circular, oval or polyhedral	Variable in shape. Fibres and sclereids
2. Cell wall	Thin cellulosic cell wall	Uneven thickening on their cell wall.	Lignified secondary cell wall present
3. Cytoplasm	Abundant	Present	Absent
4. Nucleus	Present, (Living tissue)	Present, (Living tissue)	Absent (Dead tissue)
5. Vacuoles	Large vacuole	Vacuolated	Absent
6 .Intercellular spaces	Present	Absent	Absent
7. Occurrence	Basically packing tissue, All soft part of plant, Pith,cortex , medullary rays	Dicot stems, petiole and beneath the epidermis. Absent in monocot and roots	Dicot hypodermis, bundle sheath, pericycle, Seed, pulp of fruits.
8. Functions	Food storage, Photosynthesis	Provide tensile strength, Mechanical support, Photosynthesis	Protection from stress and strain, Mechanical strength

Q. All are parts of xylem except:

A. Companion cells
B. Tracheid
C. Parenchyma
D. Xylem vessel

Q. Lignin is present in cell wall of which tissue:

a. Parenchyma
b. Sclerenchyma
c. Collenchyma
d. None

Q. Specialised cell of parenchyma present in aquatic plants to provide them buoyancy

a. Chlorenchyma
b. Sieve tube
c. Aerenchyma
d. Companion cells

Q. Which statement is incorrect with reference to cardiac tissue:

1. They are striated
2. They are voluntary in nature
3. They have features of both skeletal and visceral tissue.

Option:

a. 1 only
b. 1 and 3.
c. 2 only.
d. All are correct

Q. ……. layer is thickest and helps in protection against water loss and further secrete a waxy substance on outer surface:

a. Sclerenchyma
b. Epidermis
c. Parenchyma
d. Suberin

Q. Site of gaseous exchange on leaf is:

a. Guard cells
b. Stomata
c. Aerenchyma
d. Chlorenchyma

Q. Muscle is connected to bone via:

a. Ligament
b. Tendon
c. Cartilage
d. None

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